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ENGR 110, T 2:15 - Dr. Krishnan

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Design Review: Reflection #1

The food catering service Bon Appetit does not only work together with Santa Clara University but also for other large organizations such as TikTok and The Chase Center which all have a large customer base that need food delivered and cooked everyday. As a result, limiting our output and providing ways for Bon Appetite to better monitor their own could result in a large amount of food and money savings. For our project, we aim to examine trends connected on a quarter by quarter, week by week and hour by hour scale. We suspect there will be daily, seasonal and yearly variation depending on holidays, new workers and new food habits. This makes it even more crucial to analyze the food trends everywhere as companies can save both time and money.

The environmental impacts of this project centers on reducing food waste on campus. By analyzing trends in food sales, and by identifying what items are popular or not, the AO can collaborate with the food supplier and adjust the purchasing patterns accordingly. This will help minimize food waste from unsold items. The socioeconomic effects can be connected to the amount of staff that is working during different times of the day. The project can inform on staffing decisions based on the peak hours and the customer flow. The staffing hours can be optimized in order to cover the busy periods and make sure that the customers have a shortened waiting time.

For this project, we are working with the SCU Access Office (AO) and their work touches many people on campus. As such there are many people who are impacted by our work such as the Bon Appetit Staff, SCU Management and the AO itself. However, our primary critical customer is Nirmal who is the lead at the AO. He is primarily responsible for everything in the AO and makes data backed decisions so he needs this platform to help him.

Nirmal interacts with his community by being responsible for reporting expenses and sales data to his higher management. As such we have a dual mandate to both make data that is easily accessible as well as professionally reportable. Nirmal intends to use it as an all

encompassing solution for understanding data from Insights. He has experience with doing so manually and so he has working knowledge of methods of transformation as well as what time schedule is reasonable for delivery. He sees Tableau as a potential solution as industry standard technology.

Ethically, this project will support responsible consumption of food and transparency from the AO. Using the data will also improve the working conditions for the staff and enhance fairness and the employment well being. Improved dining experience will also contribute to better eating habits and reduced stress as the longer waiting times might lead to other food options or skipping meals. Culturally, the project will also benefit a shift in how food is valued in our community at Santa Clara as the AO will promote sustainability and responsible food choices.

Our project has a clear value to our critical customer, Nirmal, and the Access Office. By developing a Tableau-based platform, we will be able to automate the process of analyzing food sales and organizing the data, that Nirmal currently manages manually. This improvement will allow him to make decisions quicker, more accurately, and allow for more strategic decisions in regards to food purchasing and labor allocation. This platform will reduce food waste by identifying underperforming menu items, which could lead to a 5–10% reduction in wasted inventory each quarter. It will also optimize staffing schedules by aligning labor with real-time customer flow patterns, which could lower labor costs by approximately 5–8%. In addition, the improved reporting features will save Nirmal several hours per week by eliminating the need for manual data collection and it will enhance the quality of the reports he submits to higher management. Our team is well-positioned to deliver this value because we have direct access to the Access Office data, expertise in Tableau and data visualization best practices, and an ongoing working relationship with Nirmal that allows for continuous feedback and alignment with his operational needs. Through our technical skills and user experience approach, we will provide a practical, scalable, and professional solution that supports both immediate improvements and long-term operational excellence for the Access Office.

In conclusion, our project will help the Access Office and Bon Appétit improve sustainability, reduce costs, and enhance the dining experience at SCU. Working closely with Nirmal and using tools like Tableau, will allow us to come up with an efficient solution.